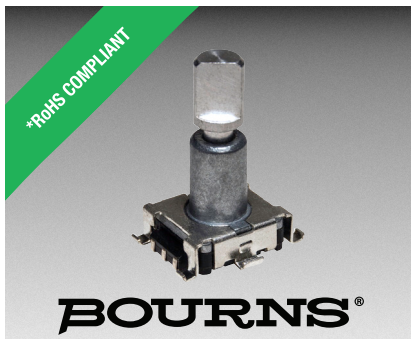




Features

- Push switch option standard
- Compact, rugged design
- High reliability
- Metal bushing/shaft
- Surface mount version
- Long rotational life

PEC11J Series – 12 mm SMD Long Life Encoder

Electrical Characteristics

Output	2-bit quadrature code
Closed Circuit Resistance	3 ohms maximum
Contact Rating	10 mA @ 5 VDC
Insulation Resistance	100 megohms @ 250 VDC
Dielectric Withstanding Voltage - Sea Level	300 VAC minimum
Electrical Travel	Continuous
Contact Bounce (15 RPM)	2.0 ms maximum
RPM (Operating)	60 maximum

Environmental Characteristics

Operating Temperature Range	-10 °C to +70 °C
Storage Temperature Range	-35 °C to +85 °C
Operating Humidity	25 % to 85 % R.H.
Rotational Life	100,000 cycles minimum
Switch Life	20,000 cycles minimum
IP Rating	IP 40
Moisture Sensitivity Level	1
ESD Classification (HBM)	N/A

Mechanical Characteristics

Mechanical Angle	360 ° continuous
Detent Torque	100 ±70 gf.cm
Shaft Push-Pull Strength	8 kg max.
Weight	5 gm maximum
Terminals	Printed circuit board surface mount terminals
Soldering Condition	
Reflow Soldering	Sn95.5/Ag2.8/Cu0.7 solder with no-clean flux (see Solder Profiles for recommended time and temperature)
Hand Soldering	Not recommended
Hardware	No hardware supplied with this encoder

Switch Characteristics

Switch Type	Contact Push ON Momentary SPST
Power Rating (Resistive Load)	10 mA at 5 V DC
Switch Travel	
S (Standard)	0.5 ± 0.3 mm
H (High)	1.5 ± 0.5 mm
Switch Actuation Force	See How to Order
Contact Resistance	100 mΩ max.

Packaging Characteristics

Pieces per Tray	75
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How To Order

PEC11J - 9 2 20 F - S 0015

Model	
Terminal Configuration	9 = Surface Mount Gull-wing Style
Detent Option	0 = No Detents (Exclusively for 24 Pulses) 1 = 18 Detents (Exclusively for 9 Pulses) 2 = 30 Detents (Exclusively for 15 Pulses)
Shaft Length	15 = 15.0 mm 20 = 20.0 mm 25 = 25.0 mm
Shaft Style	F = Flatted, Metal
Switch Configuration (SPST)	S = 350 ± 100 gf H = 400 ± 200 gf N = No Switch
Resolution	0009 = 9 Pulses per 360 ° Rotation 0015 = 15 Pulses per 360 ° Rotation 0024 = 24 Pulses per 360 ° Rotation**

** Only available without detents.

Additional Information

Click these links for more information:



CALIFORNIA WARNING: Can expose you to lead, a carcinogen and reproductive toxicant. See www.P65Warnings.ca.gov

*RoHS Directive 2015/863, Mar 31, 2015 and Annex. Specifications are subject to change without notice. Users should verify actual device performance in their specific applications. The products described herein and this document are subject to specific legal disclaimers as set forth on the last page of this document, and at www.bourns.com/docs/legal/disclaimer.pdf.

Applications

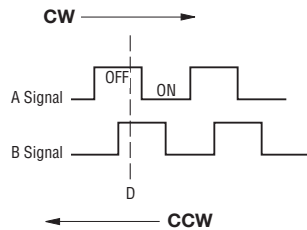
Level control, tuning and timer settings in:

- Professional audio equipment
- Professional lighting consoles
- Consumer white goods
- Test and measurement equipment
- Communications equipment
- Laboratory equipment

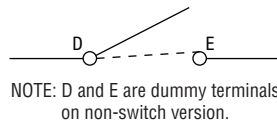
PEC11J Series – 12 mm SMD Long Life Encoder

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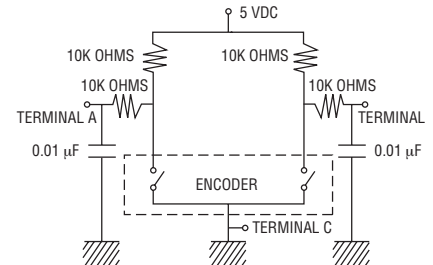
Quadrature Output Table



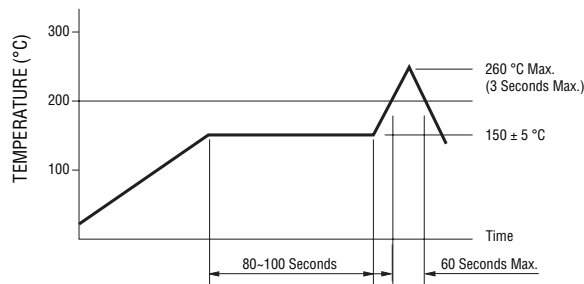
Switch Circuit



Suggested Filter Circuit



Solder Profile



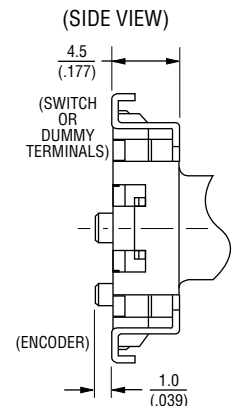
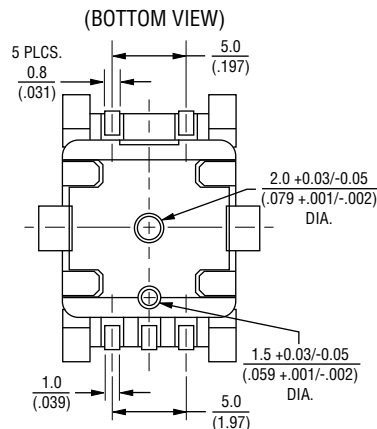
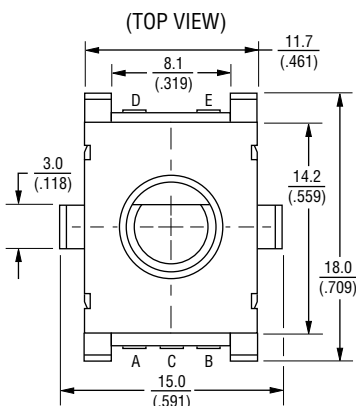
Processing Method: Reflow soldering with infrared heat or forced air convection (only once).

Notes:

1. No-clean solder paste is recommended.
2. Aqueous wash is not recommended.
3. Use of water soluble soldering flux should be avoided due to possible corrosion.
4. Multiple passes through the soldering process is not recommended.

Product Dimensions

Common Dimensions



DIMENSIONS: $\frac{\text{MM}}{(\text{INCHES})}$

TOLERANCE: $\frac{<10}{(<.400)} = \frac{\pm 0.3}{(\pm .012)}$
 $\frac{\geq 10}{(\geq .400)} = \frac{\pm 0.5}{(\pm .020)}$

Specifications are subject to change without notice.

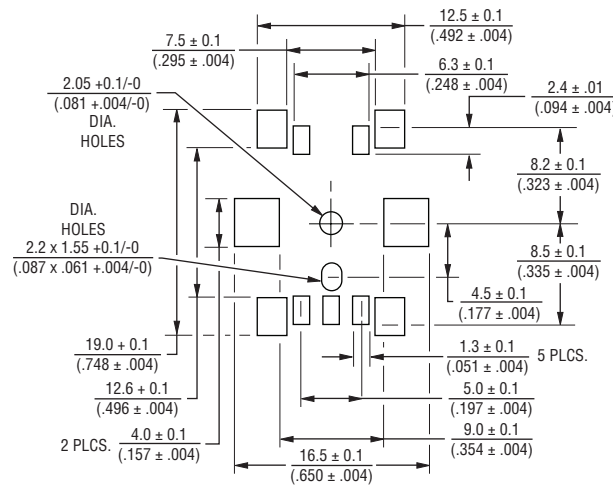
Users should verify actual device performance in their specific applications.

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PEC11J Series – 12 mm SMD Long Life Encoder

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Recommended PCB Layout



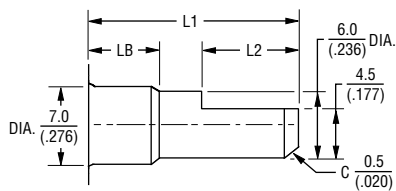
DIMENSIONS: $\frac{\text{MM}}{(\text{INCHES})}$

TOLERANCE: $\frac{<10}{(<.400)} = \frac{\pm 0.3}{(\pm .012)}$
 $\frac{\geq 10}{(\geq .400)} = \frac{\pm 0.5}{(\pm .020)}$

Product Dimensions - Shaft Styles

PEC11J without Switch

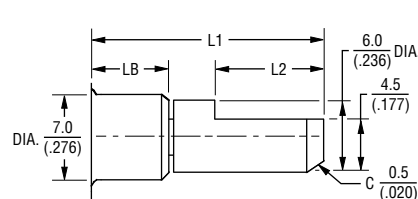
FLATTED SHAFT



L1	LB	L2
15 (.591)	7 (.276)	5 (.197)
20 (.787)	10 (.394)	7 (.276)
25 (.984)	10 (.394)	10 (.394)

PEC11J with 0.5 mm Switch Travel

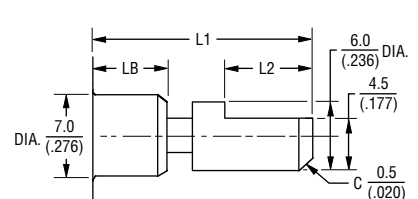
FLATTED SHAFT



L1	LB	L2
15 (.591)	7 (.276)	5 (.197)
20 (.787)	10 (.394)	7 (.276)
25 (.984)	10 (.394)	10 (.394)

PEC11J with 1.5 mm Switch Travel

FLATTED SHAFT



L1	LB	L2
15 (.591)	7 (.276)	5 (.197)
20 (.787)	10 (.394)	7 (.276)
25 (.984)	10 (.394)	10 (.394)

REV. 06/26

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